

ichosynth — Usage Manual

How to play the sampler-sequencer you *draw*

You draw music on a 16×16 LED grid with **3 knobs**. No computer, no menus to memorize: turn, press, listen.

You don't need a computer to play: your **ichosynth** generates everything on its own. Plug in your headphones, power it via USB, and off you go.

FORK

This is the 3-encoder build. Compared to the original 4-knob NI404, the **volume** is set with the **LEFT** knob, the **BPM** with the **CENTER** one, and the commands the original put on the 4th encoder (Play/Pause, Volume/BPM, Menu, Note-Shift) have been **remapped** onto the 3 available buttons. New in the fork: optional **OLED** screen and **MIDI clock OUT** (master sync).

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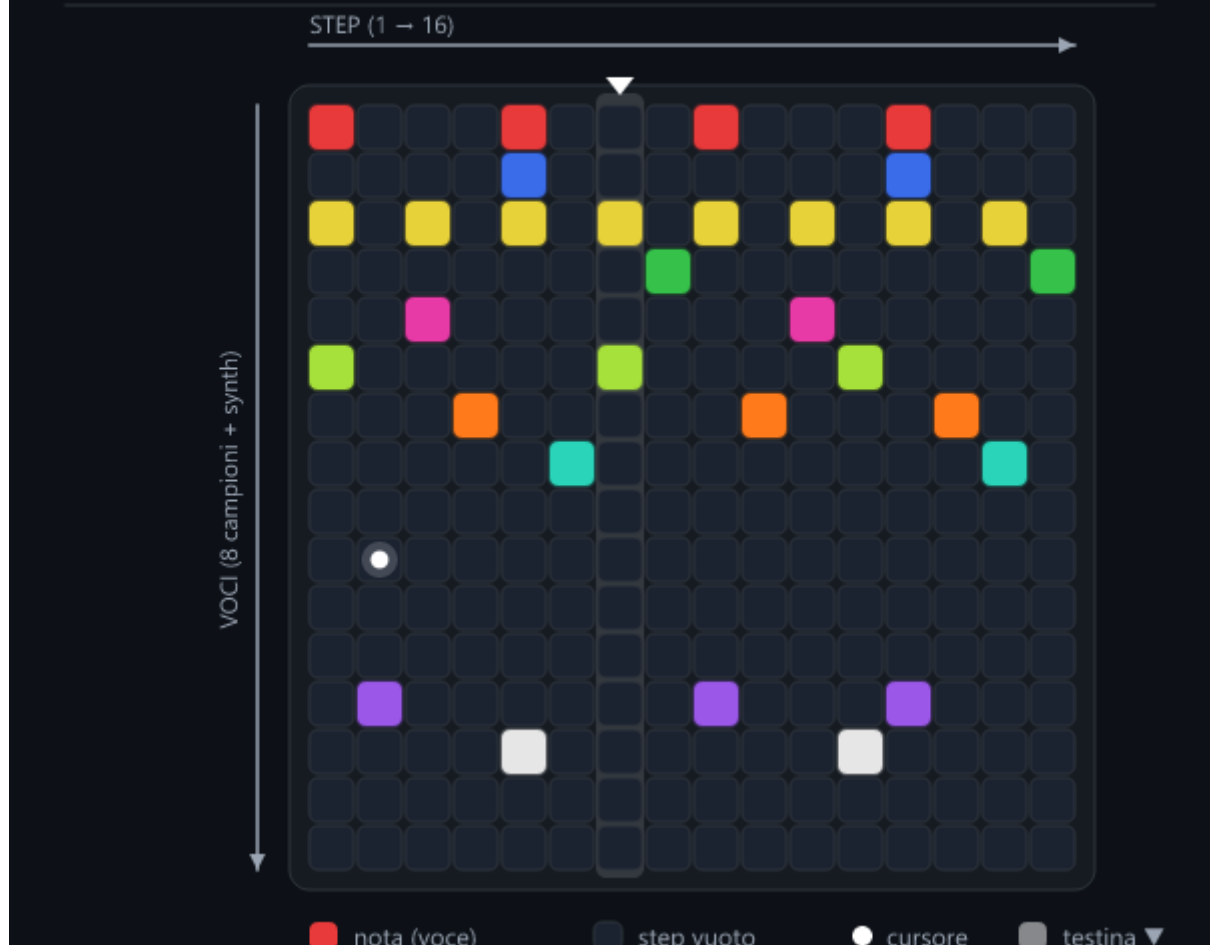
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1 · The concept in 30 seconds

The **16×16 grid** is your music sheet. A playback "playhead" runs from left to right: every column it touches plays the notes you've put there.

La griglia 16×16 = il tuo foglio musicale

Righe = voci (colori) · Colonne = step · la testina suona colonna per colonna



- The **columns** (left->right) are the **16 steps** of one bar.
- Each **row** is a **voice** (a sample or a synth), identified by a **color**.
- Several pages in a row make up a **pattern/song**.

TIP

Basic flow: **draw notes** -> **press Play** -> **loop**. You change samples, BPM and volume on the fly, without stopping.

2 · The 3 knobs (encoders)

There are **3 knobs**: **LEFT (L)**, **CENTER (C)** and **RIGHT (R)**. Each one **turns** (moves the cursor or adjusts values) and **presses**. It recognizes different gestures: single *click*, *double-click*, *hold* (long press) and *push* (press and keep held).

Le 3 manopole (encoder): gira e premi

Versione a 3 encoder. Ogni manopola si gira (cursore/valori) e si preme (azioni).



SINISTRA

cursore $\updownarrow$ · Volume

click → cancella nota

hold → cancella continuo

2× click → Single



CENTRALE

pagina · BPM

push → disegna · hold → paint

2× click → Play / Pausa

click → indietro/Vol-BPM



DESTRA

cursore $\leftarrow\rightarrow$

click → mute voce

hold → mute tutto

2× click → velocity

Knob	Turning	Pressing (main functions)
LEFT (L)	cursor up/down (Y) · in Volume/BPM: sets the Volume	click = delete note · double-click = Single mode
CENTER (C)	selects the page · in Volume/BPM: sets the BPM	push = draw note · hold = continuous drawing · double-click = Play/Pause · click = back/exit
RIGHT (R)	cursor left/right (X)	click = mute voice · hold = mute all · double-click = velocity

The **cursor** is the pulsing white dot: it shows where you're about to act.

TIP

The **CENTER** knob is the "richest" one: turn it for the page, **press** (push) to draw, **hold** to draw continuously, **double-click** for Play/Pause, **single click** to go back from menus.

3 · Reading the grid

- **Rows** = the voices (up to 8 samples + synth voices), each with its own **color** (see [ch. 11](#)).
- **Columns** = the 16 steps of the current page.
- **Top row (status)**: on the left the **8 pages** (indicators), on the right the status lights (copy

active, etc.). During Play the page indicators turn **green**.

- **Play playhead**: the highlighted column that advances as you play (the in the image above).

FORK

If you've fitted the fork's **OLED**, there you can read in plain text: mode, BPM, volume, velocity, page and Play/Stop state.

4 · DRAW mode (drawing)

This is the main screen, the default one, where you create patterns.

Action	Gesture
Move the cursor	turn L (up/down) and R (left/right)
Add a note	push C (center) on the cursor's spot (you hear the sound right away). Pressing again on a note changes the voice (cycles through them)
Continuous drawing (Etch-A-Sketch)	hold C to enable <i>paint mode</i> , then move the knobs to draw a trail of notes. Release/click to stop
Delete	click L = delete the note under the cursor · hold L = continuous deletion
Play / Pause	double-click C (center)

5 · Pages and patterns

- The grid shows **one page** (16 steps) at a time.
- Turn **C** to change page (up to **8 pages** -> **128 steps** per song).
- In Play, the pages with notes are played in sequence on a loop.

Page action	Gesture
Copy / paste page	click L + click R together: copy; repeat on another page to paste
Clear the whole page	hold L + hold C together

6 · Mute (silencing voices)

- **Click R**: mutes/unmutes the **current voice** (the row you're on).
- **Hold R**: mutes **everything** while you keep it held (release to reactivate). Perfect for live breaks/drops.

7 · Volume and BPM

- **Hold R + hold C** (together): opens the **Volume/BPM** screen.
- Inside: **Volume** = turn **L** · **BPM** = turn **C** (range ~40–240).
- To exit: **click C** (you return to DRAW).

TIP

Reminder for the 3-encoder build: there's no 4th knob, so the volume has been "moved" onto the **LEFT** and the BPM onto the **CENTER**.

8 · Velocity

- **Double-click R**: opens velocity mode for the note under the cursor.
- Adjust with **C** (turn the center knob).
- Exit by **releasing**.

9 · SINGLE mode (one voice only)

Useful for working in detail on a single sample (e.g. melodies across several pitches).

- **Enter**: **double-click L** on the row of the voice you want to isolate.
- **Exit**: **double-click L** again.

- In SINGLE the same drawing/deletion gestures from DRAW apply, but you only act on the selected voice.

Moving notes (Note Shift, in SINGLE)

- **Click R + hold C**: enters Note Shift.
- Move the notes with the knobs.
- **Click C** = confirm · **click L** = cancel.

10 · Changing the sample (Sample Browser)

To assign a different WAV file to a voice:

1. Enter **SINGLE** on the desired voice ([ch. 9](#)).
2. **Hold L + hold R**: opens the **sample browser** (SET_WAV).
3. Navigate with the knob: change **folder** and **sample** (browse the SD's files); see the waveform / length preview.
4. **Click L** = loads the selected sample onto the voice.
5. **Click C** = exit.

FILE

The samples are read from the SD in the structure /samples/<folder>/_<number>.wav (see [build manual, ch. 9](#)).

11 · Voice colors

Each voice has a fixed color (defined in [colors.h](#)):

Legenda colori delle voci

Ogni riga della griglia è una voce, identificata da un colore (da colors.h).

	Voce 1	rosso	voce campione
	Voce 2	blu	voce campione
	Voce 3	giallo	voce campione
	Voce 4	verde	voce campione
	Voce 5	magenta	voce campione
	Voce 6	verde-lime	voce campione
	Voce 7	arancione	voce campione
	Voce 8	turchese	voce campione
	Voce 13	viola	synth (sawtooth)
	Voce 14	bianco	synth (square)

The two **synth** voices (13 and 14) play internally generated waves (*sawtooth* and *square*) and follow the pitches of a scale (C3...D5).

12 · Sample Pack (sample set)

A "sample pack" is a complete set of voices saved on the SD: you recall a whole kit on the fly.

- In DRAW: **hold L + hold R** opens the **Sample Pack** screen.
- Select the pack number with the knob.
- **Click R** = saves the current set into that pack · **Click L** = loads the pack · **Click C** = exit.

FILE

On the SD a pack is the numbered folder 1..99 containing 1.wav..12.wav (ichosynth manages them, you don't need to create them by hand).

13 · Saving and loading your songs

- In DRAW: **click L + hold C** opens the **Menu**.
- **Click R** = **saves** the song into the current slot · **Click L** = **loads** · **Click C** = exit.
- Songs are saved to the SD's root as <number>.txt (up to 100).

SALVATAGGIO

Autosave/Autoload: ichosynth saves automatically to autosaved.txt (e.g. when you pause) and reloads that content at power-on — you find your work right where you left it.

14 · MIDI

All MIDI goes through the Teensy's **USB port** (you need `USB Type = Serial + MIDI` at compile time).

- **MIDI In (USB)**: ichosynth receives notes (mapping them onto the grid/current voice) and **syncs** to external MIDI clock/start/stop (slave).

- **MIDI Clock Out (USB)** (*fork, if `MIDI_CLOCK_OUT_ENABLED = 1`*): when you play, ichosynth sends **Clock (24 PPQN), Start and Stop**, so external instruments sync to ichosynth (master).



ATTENZIONE

The 5-pin DIN MIDI jacks (pins 0/1) are provided in the wiring but the current firmware does **not** use them: MIDI works only over USB.

15 · Lowpass filter (fork)

Fork feature (if `FILTER_ENABLED = 1`, on by default). Adds a **per-voice lowpass filter**: each voice gets its own cutoff that softens or "closes" the sound (rolls off the highs).

It needs one extra **momentary pushbutton** (see the build manual): a plain button between **pin 41** and **GND**. It's not an encoder, just a button — the 3-knob hardware stays as it is.

How to use it:

1. Move the cursor onto the **voice** you want to filter (the row, just like for mute).
2. **Hold the FILTER button** and **turn the CENTER knob**:
 - clockwise = filter **open** (bright sound, up to 9 kHz);
 - counter-clockwise = filter **closed** (dark/muffled, down to ~280 Hz).
3. A **light bar** across the middle of the grid shows the cutoff, in the voice's color. **Release** to exit.

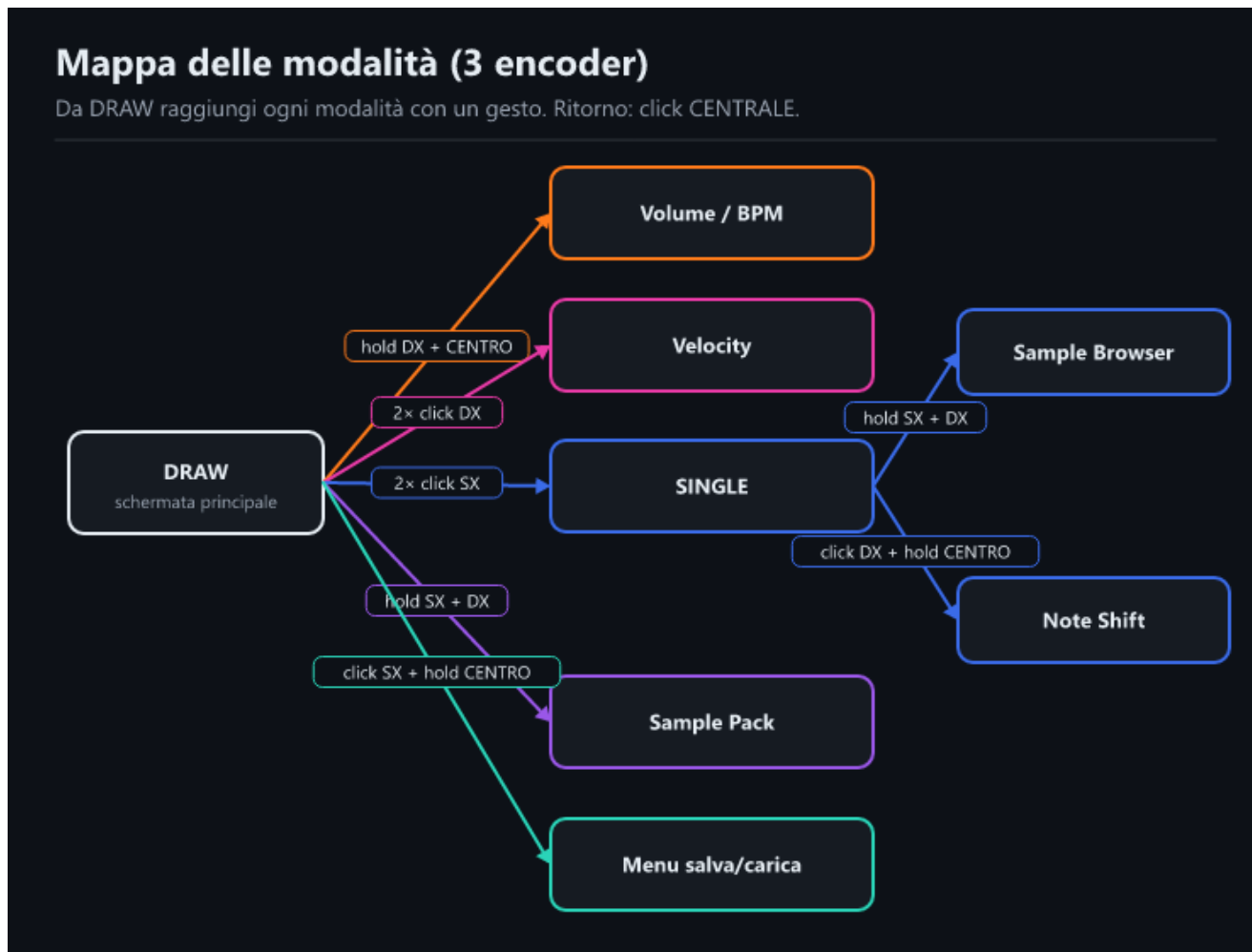
The value is **per voice** and is **saved with the song** (it comes back identical on reload).

TIP

At power-on the filters start **open** (9 kHz): the dry sound is brighter than the original NI404 (which left the filters at a ~1 kHz default, slightly muffled). For behavior identical to the original, set `FILTER_ENABLED 0` in `config.h` (and skip the button).

16 · Mode & command map

From DRAW (the main screen) you reach all the other modes with combinations of gestures:



Legend: **L** = left · **C** = center · **R** = right · "click" = short press · "hold" = long press · "push" = press and keep held · "2× click" = double-click.

You want to...	Gesture
Move cursor up/down	turn L
Move cursor left/right	turn R
Add a note	push C
Continuous drawing (Etch-A-Sketch)	hold C , then move
Delete a note	click L
Continuous deletion	hold L , then move
Play / Pause	double-click C
Mute current voice	click R

You want to...	Gesture
Mute all (momentary)	hold R
Change page	turn C
Copy/paste page	click L + click R
Clear the page	hold L + hold C
Volume / BPM (enter)	hold R + hold C -> Vol = turn L , BPM = turn C
Volume / BPM (exit)	click C
Note velocity	double-click R (adjust with C)
Lowpass filter (current voice)	hold the FILTER button + turn C
Enter/exit Single mode	double-click L
Note Shift (in Single)	click R + hold C
Sample browser (in Single)	hold L + hold R
Sample Pack (in Draw)	hold L + hold R
Save/load menu (in Draw)	click L + hold C

ATTENZIONE

Watch out for two similar gestures: *Clear page* = **hold L + hold C**; *Save/load menu* = **click L + hold C**. Only L changes (keep held vs tap).

17 · Common problems

Symptom	Fix
I don't hear anything	check the volume (hold R + hold C, then turn L), that the voice isn't muted (click R), and that the sample exists on the SD in the right format
It won't start / won't go to Play	Play is a double-click on the CENTER (two quick taps), not a single click
A knob goes the wrong way	it's a CLK/DT swap from the wiring stage (see the build manual)
The samples won't play	wrong SD path/structure, or WAV not mono/16-bit/44.1 kHz -> use wavmaker . py
I lost my work	there's the autosave (autosaved . txt), but also save often manually into a slot from the Menu
Maximum sample length	~30 seconds, with continuous looping

Have fun!

*ichosynth is a fork (3-encoder build) of **NI404** by SP_ (soundpauli) · open-source MIT firmware.*

This firmware is declared by the author "far from complete": some functions (filters, a few shortcuts) are partial. You're free to modify and improve it as you like.